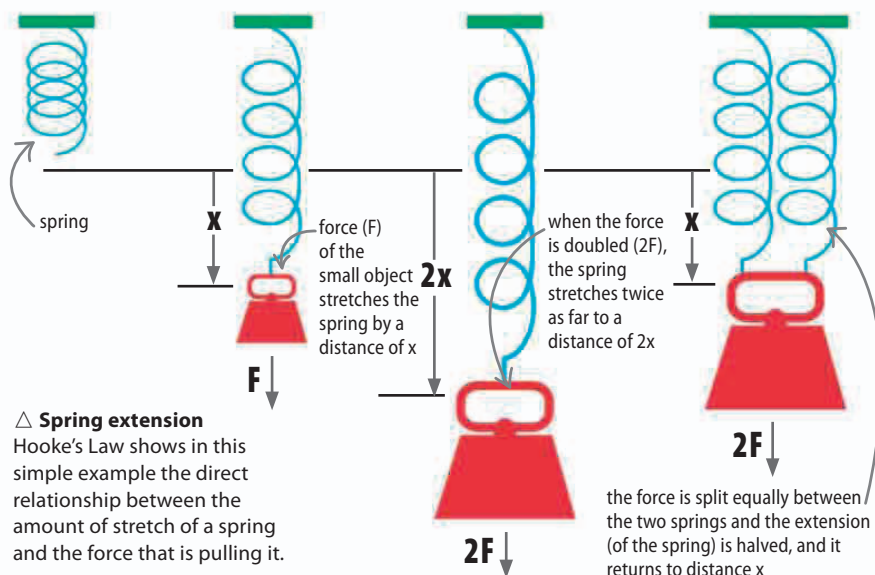


Hooke's Law

The English scientist Robert Hooke (1635–1703) discovered the law of elasticity. Hooke's Law states that the amount of stretch of a spring, or other stretchy object, is directly proportional to the force acting on it. The law is only true if the elastic limit of the spring has not been reached. If the elastic limit has been reached, the spring will not return to its original shape and may eventually break.



△ Spring extension

Hooke's Law shows in this simple example the direct relationship between the amount of stretch of a spring and the force that is pulling it.

REAL WORLD

Bungee jump

If you fell a long distance while attached to a rope, you would stop suddenly, with a dangerous jerk. Elastic cords, such as those used by bungee jumpers, slow you down more gradually because the energy of the fall is slowly transferred to the cord as it stretches.



Young's modulus

The elasticity of an object depends on its shape, size, and structure. The English polymath Thomas Young (1773–1829) devised a way of measuring the elasticity of solids—known as Young's modulus—to compare different materials.

Stiffness of select materials

rubber	0.01–0.1
nylon	3
oak	11
gold	78
glass	80
stainless steel	215.3

△ Measuring stiffness

Young's modulus is measured in gigapascals (GPa). The higher the number, the stiffer (less elastic) the material.

Material properties

Many properties of materials are related to the way they deform under stress. They depend partly on the molecules from which materials are made, but also on the shapes and sizes of larger structures inside the material, such as crystals or fibers.

Description of materials under stress

hard	difficult to scratch or dent
tough	difficult to break or deform
plastic	changes shape permanently when stressed
elastic	returns to original size and shape when stress is removed
brittle	breaks suddenly under stress, with little deformation
ductile	can be drawn out into a wire
malleable	can be hammered into shape

△ Describing materials

These terms are used to describe the behavior of materials under stress. Many materials change their behaviors with temperature. For example, warm rubber is very elastic, but very cold rubber is brittle.